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Modelling climate change impacts on viticultural yield, phenology and stress conditions in Europe

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Abstract

Viticulture is a key socio-economic sector in Europe. Owing to the strong sensitivity of grapevines to atmospheric factors, climate change may represent an important challenge for this sector. This study analyses viticultural suitability, yield, phenology, and water and nitrogen stress indices in Europe, for present climates (1980–2005) and future (2041–2070) climate change scenarios (RCP4.5 and 8.5). The STICS crop model is coupled with climate, soil and terrain databases, also taking into account CO_2 physiological effects, and simulations are validated against observational data sets. A clear agreement between simulated and observed phenology, leaf area index, yield and water and nitrogen stress indices, including the spatial differences throughout Europe, is shown. The projected changes highlight an extension of the climatic suitability for grapevines up to 55°N, which may represent the emergence of new winemaking regions. Despite strong regional heterogeneity, mean phenological timings (budburst, flowering, veraison and harvest) are projected to undergo significant advancements (e.g. budburst/harvest can be >1 month earlier), with implications also in the corresponding phenophase intervals. Enhanced dryness throughout Europe is also projected, with severe water stress over several regions in southern regions (e.g. southern Iberia and Italy), locally reducing yield and leaf area. Increased atmospheric CO_2 partially offsets dryness effects, promoting yield and leaf area index increases in central/northern Europe. Future biomass changes may lead to modifications in nitrogen demands, with higher stress in northern/central Europe and weaker stress in southern Europe. These findings are critical decision support systems for stakeholders from the European winemaking sector.

Keywords: climate change, crop modelling, EURO-CORDEX, Europe, grapevine, representative concentration pathways, STICS, viticulture

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Introduction

In Europe, viticulture is largely recognized as one of the most important crops, playing a key socio-economic role for many countries. This continent, with the largest wine production and vineyard area in the world, is home to some of the most important and renowned winemaking regions and wines (OIV, 2013). These regions present very specific environmental conditions, required to attain high production levels and quality attributes (Schultz, 2000; Jones, 2006). In fact, climate has a predominant role on grapevine physiology (Keller, 2010), influencing its growth (e.g. Van Leeuwen *et al*., 2004; Fraga *et al*., 2014) and phenological development (Jones & Davis, 2000). Previous research has shown that climatic factors are the main drivers of vineyard yield (Santos *et al*., 2011; Bock *et al*., 2013) and wine quality (Jones *et al*., 2005b; Orduna, 2010).

In this context, climate change will become an unavoidable challenge that viticulturists will have to cope with in the future. According to the *International Panel on Climate Change* (IPCC) latest report, following different representative concentration pathways (RCP), global temperature is expected to rise between 1 °C (least severe scenario) and 5 °C (most severe scenario), over the 21st century (IPCC, 2013). Moreover, the precipitation patterns over Europe are also expected to change under a higher anthropogenic greenhouse gas (GHG) forcing, leading to drier conditions in southern Europe. As such, the whole winemaking sector is expected to be largely influenced by climate change. These projections are expected to bring heterogeneous effects on European viticulture. In the Mediterranean-like climatic areas of southern Europe, increased warming and dryness should result in an overall loss of viticultural suitability, while in central and northern

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Europe, warming conditions should improve wine-growing conditions (Moriondo *et al*., 2013; Fraga *et al*., 2016a). Therefore, the assessment of climate change impacts on this important crop may help adapting the sector to future conditions.

Climate change studies applied to viticulture usually rely on local-to-regional impact assessment, that is climate change impacts on a given viticultural region (e.g. Duchene & Schneider, 2005; Bock *et al*., 2011; Tomasi *et al*., 2011; Ruml *et al*., 2012; Koufos *et al*., 2014), or on the macroscale analysis of the bioclimatic change (Hannah *et al*., 2013; Moriondo *et al*., 2013; Toth & Vegvari, 2016). Although several studies have addressed the impacts of climate change on regional grapevine production and phenology, mostly based on statistical approaches (Webb *et al*., 2007, 2011; Malheiro *et al*., 2013; Santos *et al*., 2013; Moriondo *et al*., 2015), a wide and comprehensive approach for all of Europe is still incipient. Furthermore, most of these studies only take into account climatic factors, while several other factors are not considered in a standard statistical approach, such as the physiological effects of higher CO_2 concentration in plants, different soil properties, run-off, radiation fluxes, vineyard slopes and orientation (Moriondo *et al*., 2015).

Dynamic crop models are becoming important decision support systems for the agricultural sector (Semenov & Doblas-Reyes, 2007; Challinor & Wheeler, 2008). Although these models can be applied to a large array of crops worldwide, they are still not widely adopted in viticulture. These models simulate plant growth and development, while integrating plant phenotype, soil profiles, weather data and management options. As they effectively take into account all these factors, these models usually outperform statistical approaches, despite generally requiring more data for calibration/validation (Moriondo *et al*., 2015). Site-specific parameters for climate, soil, plants and crop management, among others, are defined as input in model runs. The simulation of crop parameters under different conditions, scenarios and abiotic stresses are key outcomes of these crop models. By coupling dynamic crop models with high-resolution climate model simulations, along with consistent soil and plant data, reliable future projections of yield, phenology and stress indicators can be generated over a wide region.

This study aimed to develop and analyse climate change projections for the viticultural sector in Europe. As such, the objectives are threefold: (i) to couple a dynamic crop model with high-resolution climatic simulations for current climates and for future climate change scenarios; (ii) to develop climate change projections for grapevine yield, phenology and stress conditions in Europe; and (iii) to discuss the impacts of climate change on the viticultural sector in Europe and possible adaptation measures.

Material and methods

Model description

Grapevine modelling was based on the crop model STICS (*Simulateur mulTIdisciplinaire pour les Cultures Standard*) (Brisson *et al*., 2008). This model, developed by the French National Institute for Agronomic Research (INRA), has been successfully applied to simulate a wide variety of crops (Coucheney *et al*., 2015) and is one of the few models that incorporates the necessary parameterizations for grapevine simulation (García De Cortazar-Atauri, 2006; Valdes-Gomez *et al*., 2009; Fraga *et al*., 2015a). Furthermore, this model has previously been used in assessing the impacts of climate change on several other crops (Courault & Ruget, 2001; Gonzalez-Camacho *et al*., 2008; Brisson & Levrault, 2010).

The model runs on a daily time-step simulating crop growth, soil water and nitrogen balances driven by daily weather data. Grapevine phenological development is described by: budburst simulated using the BRIN model (García De Cortazar-Atauri *et al*., 2009a); flowering and veraison simulated using growing degree days (base temperature of 10 °C). Temperature thresholds (i.e. frost, heat shock) are also taken into account for growth and development. To simulate biomass growth, nitrogen and carbon reserves are used, also taking into account competition between vegetative and generative organs. CO_2 effects on plant physiology and radiation use efficiency are also simulated (Brisson *et al*., 2008). Fruit growth is described by the dynamics of dry matter accumulation and water content (García De Cortazar-Atauri *et al*., 2009b). The main model components (submodels) are described in Table S1, as proposed by Moriondo *et al*. (2015). The model also considers plot characteristics concerning soil profile (e.g. field capacity, wilting point and bulk density), technical parameters (e.g. plant density, canopy geometry and fruit load), variety information and weather daily data. All these parameters were used as model inputs and are described in the subsequent sections. The model outputs collected were phenology (budburst, flowering, veraison and harvest), yield, leaf area index (LAI) and water/nitrogen stress indicators.

Current climate data

STICS requires daily climatic records as input. The necessary daily atmospheric variables include maximum air temperature (T_{max}), minimum air temperature (T_{min}), solar radiation (Rad), total precipitation (Prec), wind speed (W_{speed}), water vapour pressure (V_p) and CO_2 levels. As obtaining homogenized historical weather station data for the complete set of variables, on a daily basis and throughout Europe, is unfeasible, they were obtained from the EURO-CORDEX (Jacob *et al*., 2014) SMHI-RCA4 (Swedish Meteorological and Hydrological Institute – Rossby Centre regional atmospheric model) evaluation run (Samuelsson *et al*., 2011). This experiment is driven

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by ERA-Interim reanalysis (Dee *et al.*, 2011) over the time period of 1980–2005. ERA-Interim is an observation-based data set acting as boundary forcing in SMHI-RCA4 representation of real-world climate over the selected period. Gridded climatic variables, at a spatial resolution of 0.125° longitude × 0.125° latitude, were extracted within the European sector, 15°W–35°E, 35–60°N (Fig. 1), and used as input for STICS. As running STICS for the whole European sector (75 200 grid boxes) and for 26 years would require nearly two million model iterations, daily climatic normals were used as model inputs (e.g. averages of 1980–2005), significantly reducing the amount of runs. However, to ensure the validity of the results, the full 26-year runs were performed for specific points, corresponding to viticultural regions in Europe (Fig. S1). This previous assessment shows that STICS performs similarly when using climatic normals compared to using annual data. Nonetheless, the 26-year run allows the assessment of the output interannual variability (Fig. S2). This measure indicates that yield and water stress show the largest interannual variability, whereas phenological timings, LAI and nitrogen stress show lower deviations (Fig. S2).

Soil and terrain data

As each grid box of the climate data set was considered as an independent site in the crop model, all other location-specific inputs (e.g. soil, terrain) are then defined by grid box. Soil data were obtained from the Harmonized World Soil Database (HWSD; FAO/IIASA/ISRIC/ISSCAS/JRC, 2012), while terrain data, such as slope degree and orientation, were obtained from the GTOPO30 digital elevation model (https://lta.cr.usgs.gov/GTOPO30). Soil properties from the HWSD were originally mapped at a 1 : 1 000 000 scale and are available from two layers (depths 0–30 cm and 30–100 cm) at a resolution of ~1 km (Table 1). The predominant soil type inside each climatic grid cell was then extracted to describe regional soil characteristics. Some soil properties for STICS were estimated using pedotransfer functions (Bouma, 1989), such as soil albedo, run-off, soil permeability, field capacity, wilting point and bulk density. Additionally, some soil properties were set as standard following Brisson *et al.* (2008): initial soil water content (set at field capacity), the maximum unimpeded root depth (200 cm), initial root density (0.05 cm cm^{-3}) and the soil organic N (6% dry soil).

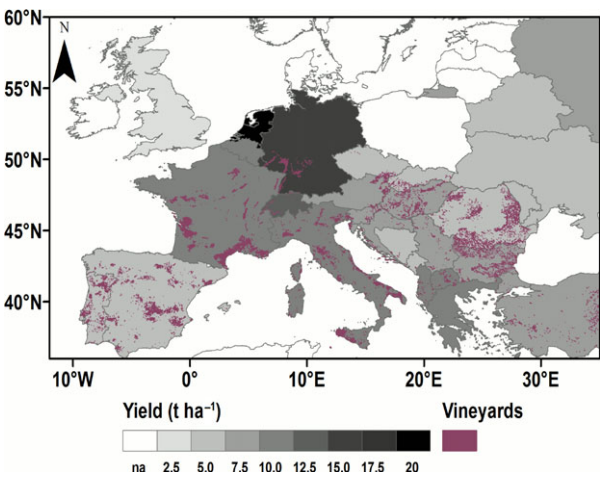


Fig. 1 Grapevine yield and vineyard area over Europe following the FAO stat (http://faostat.fao.org/) and CORINE Land Cover data sets (EEA, 2002).

Table 1 Soil and terrain parameters as defined by STICS, along with the corresponding data sets used for their calculation and key references

Parameter	STICS parameter	Calculation
Clay content in the surface layer (%)	Argi	HWSD
Limestone content in the surface (%)	Calc	HWSD
Soil pH	pH	HWSD
Albedo of the dry soil	albedo	Beven, (1995)
Cumulative evaporation limit (mm)	q0	Ringrose-Voase *et al.*, (2003)
Fraction of run-off in soil	ruisolnu	Chapman & Lake, (2003)
Dominant soil unit (FAO)	*	HWSD
Permeability classes	*	Ritchie, (1985)
Texture class	*	HWSD
USDA Texture	*	HWSD
Mean slope (%)	*	GTOPO30
Orientation	*	GTOPO30

HWSD, Harmonized World Soil Database; STICS, Simulateur mulTIdisciplinaire pour les Cultures Standard.
Parameters used in pedotransfer functions are highlighted by *

Grapevine varieties and cultural practices

For a large-scale comprehensive modelling approach throughout Europe, some assumptions were made concerning grown varieties and cultural practices. Not being possible to define varieties grown at each location, a standard variety was considered for all locations. For a European-wide study, the choice of the variety is of utmost importance. The selected variety was Pinot noir, due to its early-to-intermediate ripening and moderate yields, being suitable for a large number of viticultural regions (Jones, 2006). Pinot noir is currently the 10th (6th red) most planted variety and the 7th fastest-expanding variety in the world (Anderson & Aryal, 2013). As it is grown in vineyards throughout Europe, from the warmer countries in southern Europe to the cooler central–northern European wine regions (Anderson & Aryal, 2013), it was selected herein.

Cultural practices were kept invariant in model runs, and crop interventions were set to a minimum (i.e. topping, thinning and leaf removal were not considered in the model). The default trellis system was Cordon, and vine density was set to 4000 vines ha^{-1}, a reasonable value considering most

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commercial vineyards in Europe (Jackson, 2008). This choice may affect some results, such as the total yield values, but may allow to obtain a first assessment of the future conditions. Irrigation and fertilization were not considered/applied. However, because grapevines are perennial crops, the N reserve available in perennial organs was considered in the model initializations. Harvest decision was set to be determined by STICS once water content of the berries reached a maximum of 77%, corresponding to an alcohol level of approximately 12.5 (%v/v) (García De Cortazar-Atauri *et al*., 2009b).

Model validation

The current study focuses on the potential areas for viticulture taking into account the whole European sector and not only areas on which viticulture is currently established. It should be noted that in some currently suitable areas for grapevines (e.g. in northern Europe), viticulture is not implemented possibly because it is not economically interesting. While applying STICS only to areas where viticulture is currently present would provide more accurate results, it would leave out the potential new viticultural areas under climate change projections. As such, a criterion was set: areas where the model provided yield results = 0 were considered inappropriate for winegrape growth. This threshold (yield = 0) is obtained when grapes do not reach full maturity. Nonetheless, a suitable model validation throughout Europe becomes necessary to ensure STICS is accurate in simulating real-world conditions. To perform this validation, model outputs were compared to observations for (i) yield and (ii) phenology. For (i), country statistics from the Food and Agriculture Organization of the United Nations (FAO; http://faostat.fao.org/) were used. For (ii), data were collected from literature from the several sources (Table 2). It should be noted that the data used for model verification correspond to a large number of varieties (mixed varieties) and years. Despite these shortcomings, this validation is still useful to assess whether STICS is able to capture the diversity of viticultural conditions throughout Europe.

Climate change projections

Climate models are the most valuable tool in climate change assessments (IPCC, 2013). In the current study, to assess climate change impacts on European viticulture, four pairs of Global Climate Model/Regional Climate Model (GCM/RCM) simulations were used (Table 3), retrieved from the EURO-CORDEX project. The use of this multimodel ensemble allows taking into account numerical model uncertainties, such as different modelling approaches, model parameterizations and initializations (Deser *et al*., 2012). These simulations integrate the Coupled Model Intercomparison Project (CMIP5) and are forced by different RCP scenarios. Data used correspond to RCP4.5 and RCP8.5 scenarios for 2005–2100. In RCP4.5, CO_2 emissions are assumed to increase until mid-21st century, decreasing afterwards (IPCC, 2013). In contrast, in RCP8.5, the CO_2 emissions continue to rise until the end of the 21st century. Data sets were submitted to model calibration and bias correction using the evaluation runs forced by ERA-Interim reanalysis data (Dee *et al*., 2011). As the number of simulations is relatively large, only the ensemble mean for 2041–2070 (as for the observed data) is shown herein to isolate the climate delta and to assess regional-scale uncertainties. CO_2 levels for the two scenarios were used as model inputs in STICS (IPCC, 2013), while for the recent past period CO_2 levels were set at 350 ppm.

Table 2 Literature sources containing phenological data for the selected countries in Europe

Country	Reference
Croatia	Jones *et al*., (2005a), Koch *et al*., (2010)
France	Jones & Davis, (2000), Barbeau, (2007), Jones *et al*., (2005a), Koch *et al*., (2010)
Germany	Nendel, (2010), Bock *et al*., (2011), Urhausen *et al*., (2011), Jones *et al*., (2005a), Koch *et al*., (2010)
Italy	Caffarra & Eccel, (2010), Marta *et al*., (2010), Tomasi *et al*., (2011), Mariani *et al*., (2013), Orlandi *et al*., (2015)
Portugal	Coelho *et al*., (2013), Malheiro *et al*., (2013), Real *et al*., (2015), Fraga *et al*., (2016b)
Slovakia	Koch *et al*., (2010)
Slovenia	Koch *et al*., (2010)
Spain	Rodríguez *et al*., (2010), Fernandez-Gonzalez *et al*., (2013), Ramos *et al*., (2015)

Table 3 List of the ensemble members used in this study, along with their respective institutions.

GCM-RCM	Institutions
CNRM-CERFACS – SMHI-RCA4	Centre National de Recherches Météorologiques
IPSL-CM5A-MR – SMHI-RCA4	Institut Pierre Simon Laplace
MOHC-HadGEM2 – SMHI-RCA4	Met Office Hadley Centre
MPI-M-MPI – SMHI-RCA4	Max Planck Institute

GCM-RCM, Global Climate Model/Regional Climate Model.

Results

Historical/evaluation simulations

Phenological timings. The modelled phenological timings for budburst, flowering, veraison and harvest under current climate (1980–2005) are shown in Fig. 2 (a, b, c and d, respectively). The overall patterns for viticultural areas (Fig. 2 colour areas) present a very high agreement with current European vineyard land

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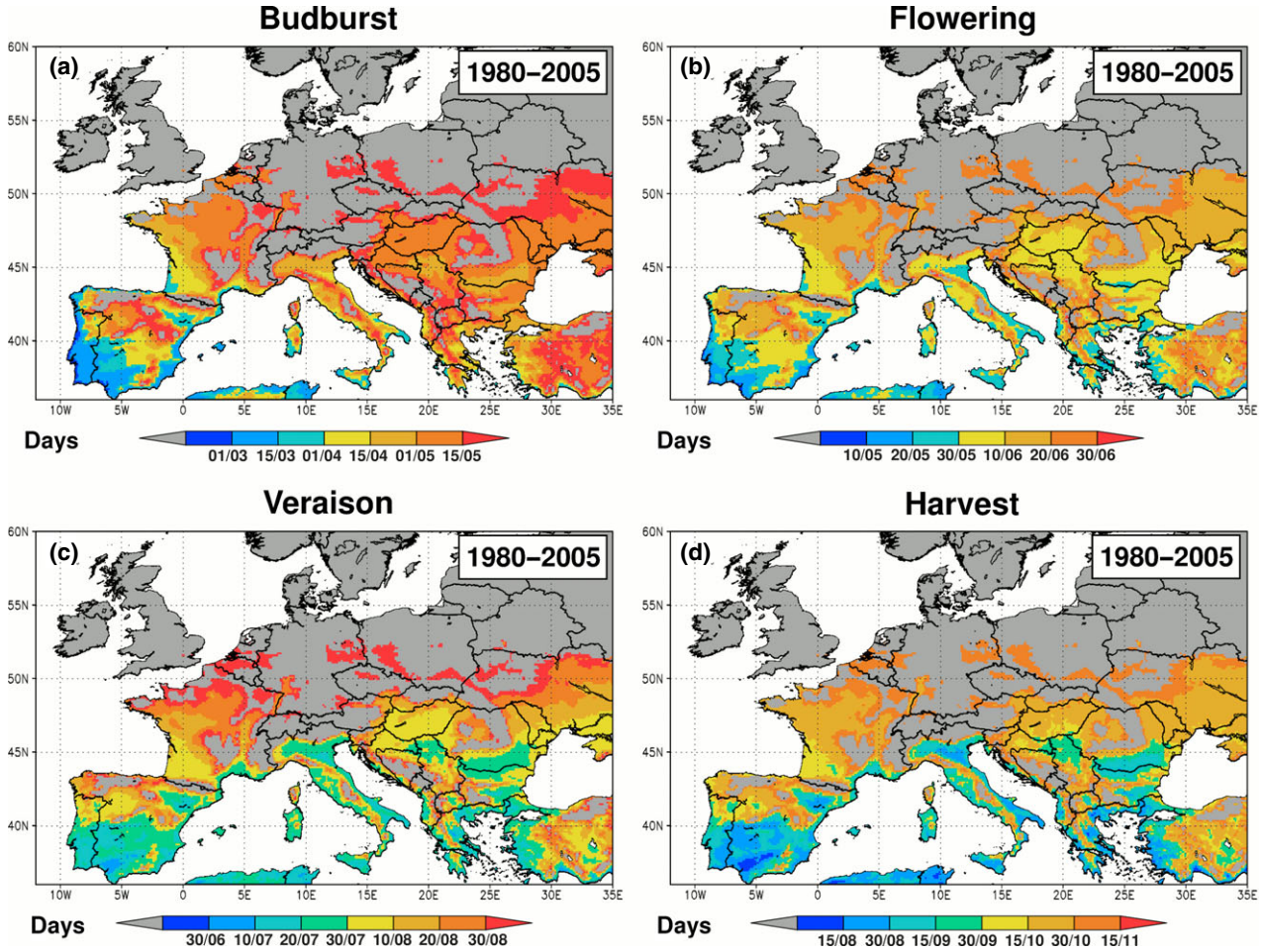


Fig. 2 Simulateur mulTIdisciplinaire pour les Cultures Standard (STICS) model simulations for the (a) budburst, (b) flowering, (c) veraison and (d) harvest timings over Europe (1980–2005). The areas where STICS provides no yield outputs (0 t ha^{-1}) are shown in grey.

cover (Fig. 1). In general, all phenological timings tend to occur earlier in southern/coastal areas of Europe as opposed to central/northern areas, which tend to occur latter. Budburst (Fig. 2a) ranges from the 2nd week of March in the Iberian Peninsula to the 2nd week of May in Germany, Poland and some areas in eastern Europe. When comparing results to observational phenology (Table 4), the model presents a high skill (r = 0.86), but with moderate error ($_{cv}$RMSE = 18%), showing an average overestimation of this event by 14 days (budburst is delayed in the model). Flowering (Fig. 2b) tends to occur from the 2nd week of May until the end of June. This event also shows a high agreement with observed phenology in Europe (r = 0.80) and a relatively low error ($_{cv}$RMSE = 4%). Veraison (Fig. 2c) is modelled to occur from the end of June to the end of August, with a moderate skill (r = 0.68) and a low error ($_{cv}$RMSE = 4%). Modelled harvest dates (Fig. 2d) occur from the end of August until the end of October. STICS also tends to overestimate this event, showing a moderate skill (r = 0.69) and an error of 10%. It should be noted that while the observational data correspond to the harvest, the modelled timings correspond to the timing when alcohol content reaches 12.5° (%v/v), which is not consensual across Europe.

Biomass production: leaf area and yield. Vineyard LAI modelled by STICS (Fig. 3a) shows an overall maximum value of 3 m^2 m^{-2} occurring during veraison, which is normal for inter-row type crops (Ollat *et al.*, 1998). Near-maximum values tend to occur only in the winemaking regions of France and northern Iberia, whereas lower values (1.5–2 m^2 m^{-2}) occur in central/western Iberia and regions in the Mediterranean and the Adriatic. Although the modelled LAI refers only to vineyards, the pattern over Europe provides a high agreement with satellite-derived LAI for all vegetation, namely the north–south European contrasts (cf. Stockli & Vidale, 2004; their Fig. 6). However, a more detailed comparison between modelled and satellite-derived

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Table 4 Comparison between the observed vs. STICS simulated phenological timings (BUD – budburst, FLO – flowering, VER – veraison and HAR – harvest) in 35 wine regions across Europe

COUNTRY	REGION	LON	LAT	BUD (obs)	BUD (sim)	FLO (obs)	FLO (sim)	VER (obs)	VER (sim)	MAT (obs)	HAR (sim)
Croatia	Croatian Uplands	17.23	45.60	110	126	160	161	226	216	262	293
	Dalmatia	17.98	42.75	93	111	148	154	218	209	252	266
	Slovania and Danube	18.23	45.37	112	123	159	158	224	213	262	292
France	Alsace	7.38	48.20	109	135	169	170	237	237	275	305
	Bordeaux	0.55	44.50	82	100	163	155	229	216	275	282
	Burgundy	1.77	46.99	107	122	164	164	220	228	268	300
	Loire Valley	0.13	47.12			170	161	235	227	258	297
	Provence	3.44	43.35			148	146				
Germany	Franconia	10.05	49.82	121	139	174	175	231	250	275	311
	Mosel	6.87	49.28	126	139	178	175				
	Rheingau	8.23	49.30	113	134	168	169	235	237	260	304
	Rheinhessen	8.13	49.92	107	135	168	170	228	239		
Italy	Piedmont	8.67	44.66	103	119	154	157	213	211		
	Trentino	12.14	45.83	101	116	155	155	218	208		
	Tuscany	11.77	43.08	99	120	153	158				
	Umbria	12.25	43.02			140	155			254	261
	Veneto	12.61	45.63	103	114	155	154	218	208		
Portugal	Alentejo	−7.56	38.38	78	67	141	142	207	195	245	237
	Douro	−7.55	41.17	83	99	143	159	205	215	260	286
	Lisboa	−9.18	39.04	76	61	142	135	213	205	260	261
	Vinhos Verdes	−8.41	41.82	86	74	147	154	211	216		
Slovakia	Južnoslovenská	18.48	47.77	101	129	157	163				
	Malokarpatská	17.30	48.33	108	134	161	167				
	Nitrianska	18.60	48.22	106	132	163	165				
	Stredoslovenská	19.95	48.22	110	137	162	170	232	231	277	304
	Tokaj	21.77	48.37	107	130	154	163				
	Východoslovenská	21.77	48.68	100	134	162	166				
Slovenia	Podravje	15.95	46.57			169	162				
	Posavje	15.70	46.02			167	163				
	Primorska	13.93	45.83			163	169				
Spain	Empordà	2.36	41.64	102	104	155	157			267	287
	Rías Baixas	−8.42	42.52	97	91	157	160	233	228	281	295
	Ribeira Sacra	−7.46	42.41	88	109	156	165	214	229	253	297
	Ribeiro	−8.10	42.30	92	87	160	158	210	221	253	288
	Ribera del Duero	−4.36	41.63	116	128	164	165	223	218	277	294
All	*r* ($_{cv}$*RMSE*)			0.86 (19%)	0.80 (4%)	0.68 (4%)	0.69 (10%)				

STICS, Simulateur mulTIdisciplinaire pour les Cultures Standard.

LAI becomes increasingly difficult. Satellite measurements cover all occurring vegetation within a given grid cell and not only grapevines. Additionally, vineyard rows are discontinuous, with low ground cover and mixed vegetation between lines (Johnson *et al*., 2003), leading to algorithm detection constraints. Nonetheless, modelled LAI values are in clear agreement with previous studies for vineyards (Ollat *et al*., 1998; Johnson *et al*., 2003). It is clear that LAI is greatly affected by water stress, as the two patterns exhibit a large spatial agreement (Fig. 3 and 4).

Yield patterns over Europe (Fig. 3b) show the same contrast between central/northern Europe (higher yields) and southern Europe (lower yields), confirming the relationship between LAI, yield and water stress conditions (Kliewer & Dokoozlian, 2005). Lower yields (<5 t ha^{-1}) are generally found in central–southern Iberia, whereas the highest productivity wine regions are located in northern Iberia, France, Germany and some areas of eastern Europe. As an example, in Spain, STICS indicates that the highest yield region is located in the north-west (Rías Baixas), which is indeed a high productivity region (Lorenzo *et al*., 2012). These patterns also show a general agreement with the country yield statistics (Fig. 1). Nonetheless, in compliance with other studies using the STICS crop model, an overestimation

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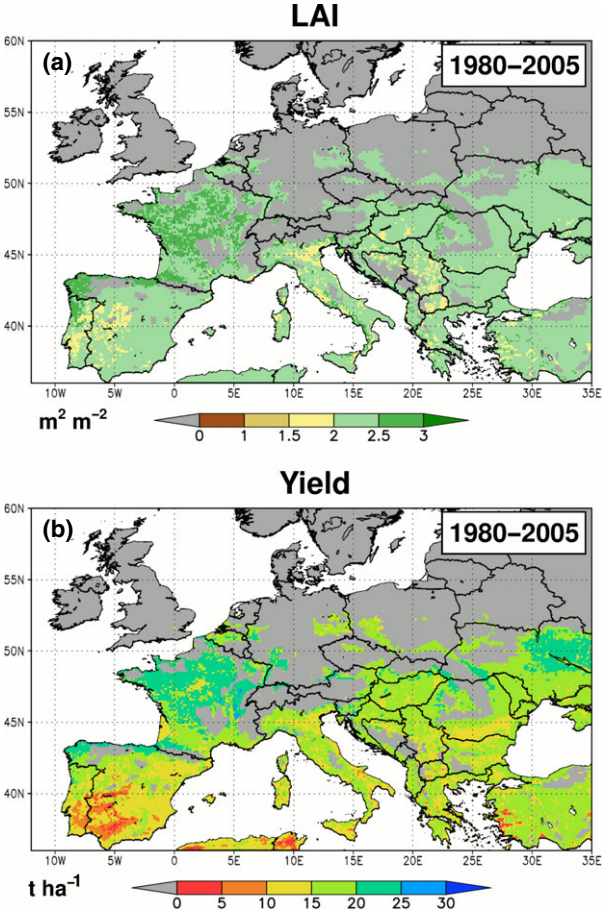


Fig. 3 Simulateur mulTIdisciplinaire pour les Cultures Standard (STICS) model simulations for (a) leaf area index (LAI) (m^2 m^{-2}) and (b) yield (t ha^{-1}) over Europe (1980–2005). The areas where STICS provides no yield outputs (0 t ha^{-1}) are shown in grey.

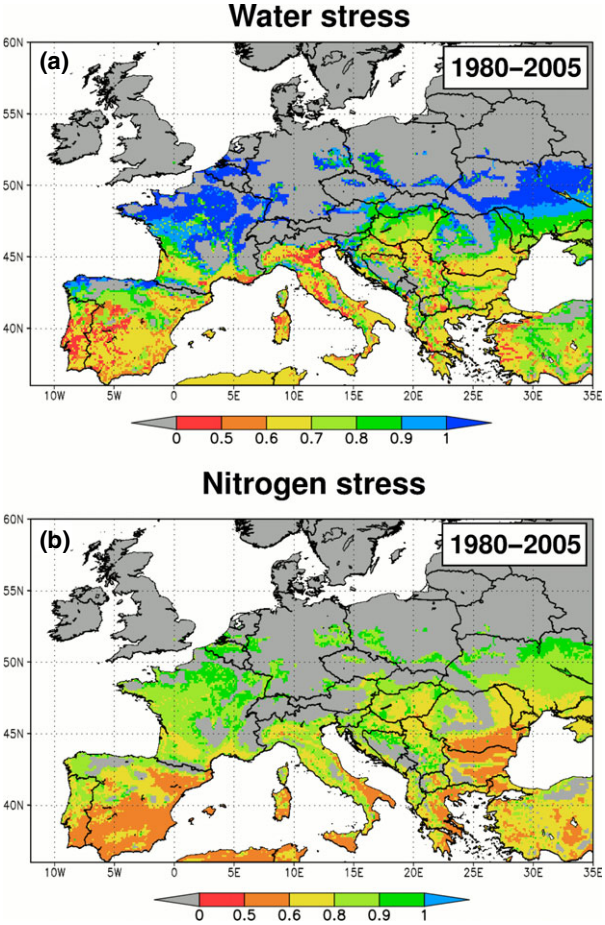


Fig. 4 Simulateur mulTIdisciplinaire pour les Cultures Standard (STICS) model simulations for the (a) water and (b) nitrogen stress indices over Europe (1980–2005). The areas where STICS provides no yield outputs (0 t ha^{-1}) are shown in grey.

of the yield values can be found (Valdes-Gomez *et al.*, 2009; Fraga *et al.*, 2015a). These discrepancies between modelled and actual yields can be attributed to the use of a single variety (Pinot noir) in the modelling approach. In reality, vineyards in Europe are composed of many rootstocks, cultivars and botanical clones determining potential yields (e.g. Renouf *et al.*, 2010). Maximum yield thresholds imposed by regulators of each winemaking region may also contribute to these differences.

Water and nitrogen stress indices. An evaluation of the STICS water and nitrogen (N) stress indices during the grapevine reproductive stage (i.e. from the start of fruit setting until maturity) was performed, as these two factors may explain the variability in yields. In STICS, the water stress index corresponds to a ratio between actual and maximum plant transpiration, where a value of 1 indicates no water stress and lower values indicate water stress, affecting growth (slower) and radiation use efficiency (decrease) (Brisson *et al.*, 2008). Consequently, this index has a high impact on LAI and yield. It should be noted that irrigation was not applied and only water supplied by precipitation was considered. Mean values of this index are shown in Fig. 4a. Overall, the pattern depicts a remarkable contrast between southern Europe, where moderate water stress already exists (<0.5), and central and eastern Europe, where no water stress is found (~1). This indicates water limitations over small areas of southern Europe, particularly in Iberia and Italy. Although it is important to notice that the assumptions made for the crop management and grapevine may have a strong impact on the water status of the plant (e.g. density, variety, training system), the results found are very similar to other studies considering water stress indicators for grapevines (Fraga *et al.*, 2013).

An assessment of the N deficiency index was also performed (Fig. 4b). Although no anthropogenic

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fertilization was considered, it is important to study N stress under future climate scenarios (Webber *et al.*, 2015). This index influences the accumulated shoot biomass of the canopy, affecting growth (slowing), leaf senescence (acceleration) and radiation use efficiency (decrease) (Brisson *et al.*, 2008). It provides the N demands by grapevines, where values lower that one depict N deficiency. Because no fertilization was applied, only the N supply from the soil to the roots is considered, which greatly depends on soil mineralization and water status (Brisson *et al.*, 2008). The N stress pattern (Fig. 4b) shows that the regions currently more affected by this problem are located in southern Europe.

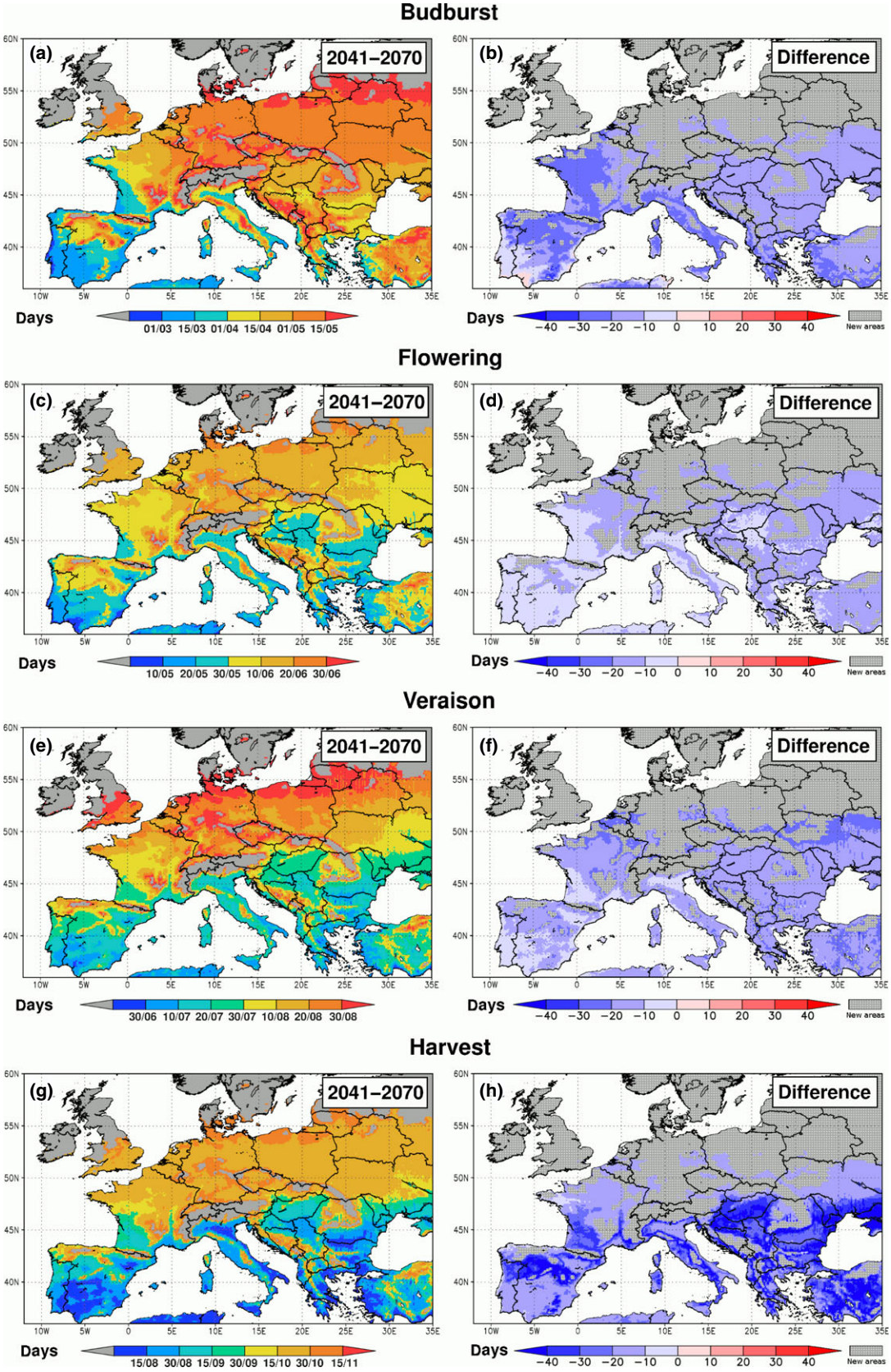
Climate change projections. An ensemble of models is used to simulate climate change projections for the selected climatic variables. Under RCP8.5, large parts of Europe are expected to be exposed to a warming of up to 4 °C until 2071 (Fig. S3a). In Great Britain, northern France and Germany, this warming should be less severe, with increases up to 2 °C. Regarding precipitation, the ensemble mean projects a significant increase in large parts of central and northern Europe and a decrease in southern Europe (Fig. S3b). The mean patterns of the climate change signal for temperature and precipitation (Fig. S3) are in agreement with a larger ensemble of models provided by the EURO-CORDEX project (Jacob *et al.*, 2014), indicating that the selected model ensemble presents relatively low bias. Despite the good agreement amongst RCMs, large uncertainties still remain regarding precipitation projections under future climates in Europe (e.g. IPCC, 2013, Knutti & Sedlacek, 2013; Santos *et al.*, 2016). Although results under RCP4.5 were also computed (Figures S5–S7), impacts on phenology, yield, LAI and water/nitrogen stress indices were very similar to RCP8.5. As such, only results under RCP8.5 will be described henceforth.

Changes in phenology. The patterns for grapevine growing areas under future climates (nongrey landmass areas in Fig. 5) suggest their northward expansion up to 55°N. Although the future northern suitability limits for viticulture are in agreement with Hannah *et al.* (2013), the southern suitability limits remain at their current levels, in opposition to that study but in agreement with Van Leeuwen *et al.* (2013). Regarding the phenology, STICS shows clear advancements in the timings of each event (Fig. 5). For the budburst timings, the climate change signal (future minus present) hints at their anticipation (Fig. 5a and b). These changes are more noticeable in regions in northern/eastern Iberia, France and Italy, with advances in the phenological timings up to 30 days. In southern/western Iberia, these differences are generally less pronounced, with advances up to 10 days. Regions in eastern Europe show changes of ca. 20 days. The anticipation of this event may also increase grower's concerns about frost risk, although recent studies suggest lower frost risks in future climates (Molitor *et al.*, 2014; Mosedale *et al.*, 2015).

Flowering reveals less pronounced projected changes that those for budburst (Fig. 5c, d). The regions with the largest anticipations are located in eastern Europe and some areas of France, as opposed to regions in western/southern Europe, where changes are up to 10 days. Projected changes in the veraison timings (Fig. 5e, f) are similar to those of flowering, but with slightly higher anticipations throughout Europe. Harvest shows the highest timing anticipations of all phenophases (Fig. 5g, h). Anticipations of up to 40 days are projected for wine regions in northern Iberia, southern France, Italy and many countries of eastern Europe (e.g. Bulgaria, Croatia, Hungary, Romania and Turkey). Smaller changes are expected for other regions in western/central Europe, such as northern France and Germany. These outcomes are in general agreement with previous studies projecting future advances in grapevine phenological timings throughout Europe (García De Cortázar-Atauri *et al.*, 2010; Donatelli *et al.*, 2012), Germany (Neumann & Matzarakis, 2011), France (Duchene *et al.*, 2010; Cuccia *et al.*, 2014), Italy (Caffarra & Eccel, 2011) and Portugal (Fraga *et al.*, 2016b). Nonetheless, the phenology submodel used in STICS does not take into account very high temperature effect (Wang and Engel model), which may result in biases in very hot regions (Cuccia *et al.*, 2014).

Due to the heterogeneous anticipation of the phenological timings, future changes in the interphase intervals might be quite diverse (Fig. S4), clearly hinting at nonlinear effects of projected warming on phenology (Sadras & Moran, 2013). For the budburst–flowering interphase (Fig. S4a), almost all of Europe may present longer periods, with the exception of some areas in coastal Portugal and southern Spain. Some areas of northern Spain and France may present much longer periods (as much as 20 days). For the flowering–veraison interphase (Fig. S4b), changes are much more homogeneous, with shortenings <10 days. For the veraison–harvest interphase (Fig. S4c), a contrast between central and southern Europe is apparent, with longer periods (up to 10 days) in the former and shortenings of 20–30 days in the latter. Due to these changes in the phenological timings/interphases, changes in the budburst–harvest period will also present some contrasts in Europe (Fig. S4d). Most of southern Europe shows shortenings, while over some central/northern

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Fig. 5 Simulateur mulTIdisciplinaire pour les Cultures Standard (STICS) model simulations for the (a) budburst, (c) flowering, (e) veraison and (g) harvest timings over Europe (2041–2070), along with the respective differences to the recent past period (b, d–f). The areas where STICS provides no yield outputs (0 t ha^{-1}) are shown in grey.

European regions the development cycle is expected to lengthen.

Changes in leaf area and yield. Future changes in grapevine leaf area in Europe are heterogeneous (Fig. 6a, b). Generally, LAI changes reflect the combined effect of increased dryness, favouring lower LAI, and increased CO_2 effect on vine physiology, favouring higher LAI (Bindi *et al.*, 1996; Moutinho-Pereira *et al.*, 2009, IPCC, 2013). Grapevine photosynthesis is stimulated by increased CO_2, leading to a biomass increase (Schultz, 2000). On the other hand, enhanced dryness may lead to growth reductions. Bindi *et al.* (1996) also confirmed this complex relationship between CO_2 levels, dryness and biomass. As a joint result of these two forcing factors, in southern European wine regions, where water stress is expected to reach limiting levels, LAI is projected to decrease. Conversely, in central/northern European wine regions, enhanced CO_2 should be the main driving factor for LAI, as water stress is expected to remain of secondary relevance. In fact, Kizildeniz *et al.* (2015) found that higher CO_2 atmospheric contents can mitigate some negative effects of drought. Nonetheless, the benefits of higher CO_2 in plants may be limited by the plant response (Ingram & Porter, 2015).

Regarding yields (Fig. 6c, d), projections indicate large decreases in productivity of up to 8 t ha^{-1} in southern Iberia (Extremadura and La Mancha in Spain and Alentejo in Portugal), in Italy (Emilia Romagna

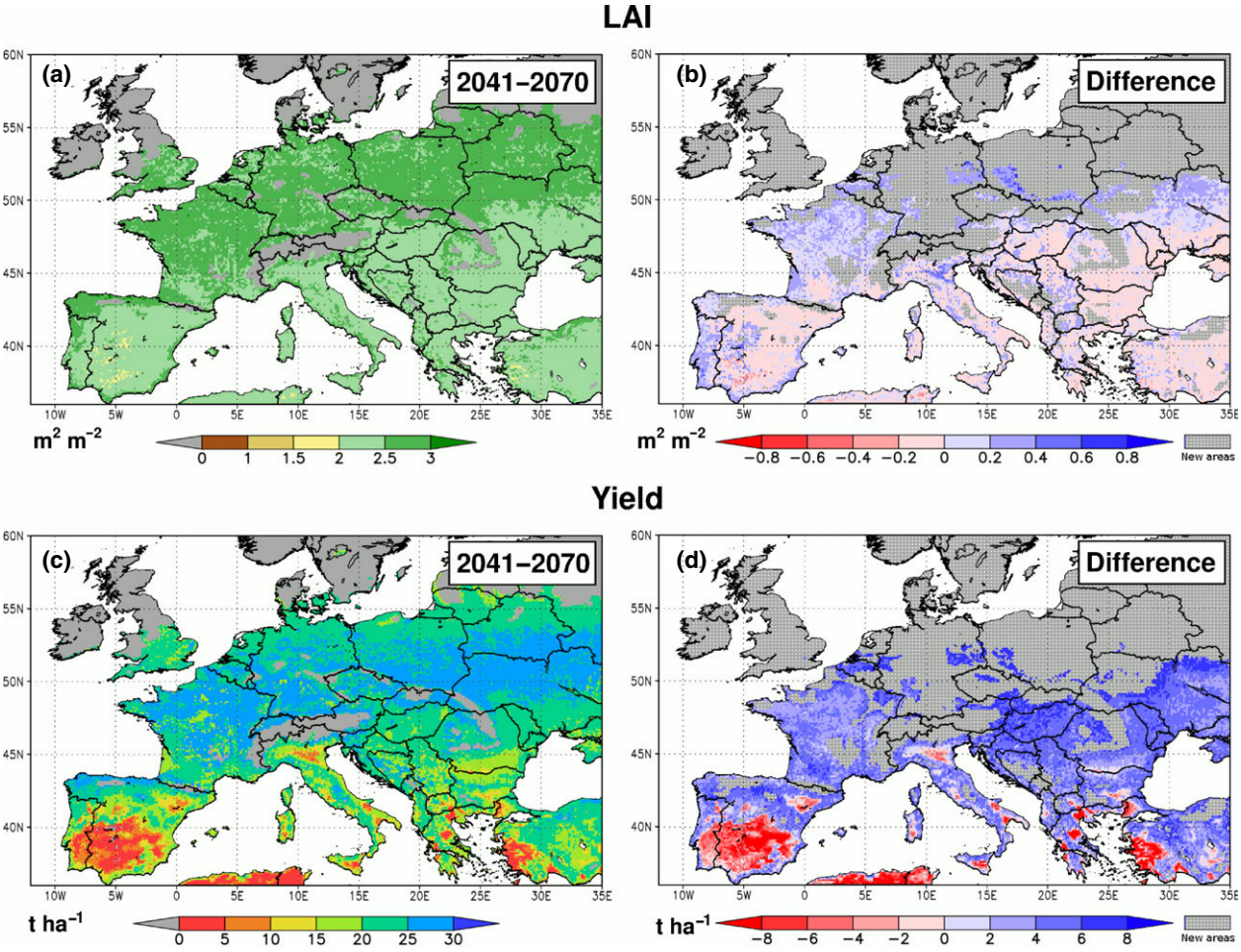


Fig. 6 Simulateur mulTIdisciplinaire pour les Cultures Standard (STICS) model simulations for (a) leaf area index (LAI) (m² m^{-2}) and (c) yield (t ha^{-1}) over Europe (2041–2070), along with the respective differences to the recent past period (b, d). The areas where STICS provides no yield outputs (0 t ha^{-1}) are shown in grey.

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and Lombardy) and along the Aegean Sea. This is certainly tied to the high water stress levels projected for these regions. In opposition, all other regions of Europe are projected to have yield increases in the future. Improved thermal conditions for grapevine growth are responsible for these increases, which are particularly evident over eastern Europe. A distinct pattern in Europe is visible in the Po Valley (Italy), where LAI increases while yield decreases. In this region, future precipitation is projected to increase in spring and decrease in summer (Jacob *et al*., 2014). This may promote plant growth at early stages, but higher summertime water stress and lower final yields. This result highlights the importance of using STICS, as it holistically integrates development timings and water status.

These results are also consistent with previous studies. For the Douro region in Portugal, Santos *et al.* (2013) projected a net increase in productivity, which is consistent with our findings for the same grid cell (western Douro valley). However, for the eastern part (hotter and dryer) of the Douro region, our projections indicate decreases in yield. For France, results are also vastly consistent with previous studies which project positive yield trends for the future (Brisson & Levrault, 2010). Bindi *et al.* (1996) projected yield increases for Bologna (Italy) under future scenarios, which is identical to our results for the corresponding grid cell. Nevertheless, Moriondo *et al*. (2011) predicted a decrease in yields in Tuscany (Italy), caused by the future shorter periods for biomass accumulation. Our results project some heterogeneities in the yield projections in that region, with both increases and decreases over the entire region. Climate projections in the previous studies are based on different RCMs and emission scenarios, which may account for the differences in the projected yields (Knutti & Sedlacek, 2013; Ferrise *et al*., 2014). Nonetheless, the large-scale data sets (climate, soils and orography) used in this study may also be a possible source of bias and a more local analysis may become necessary.

Changes in water and nitrogen stress indices. Climate change projections indicate that water stress conditions may intensify (Fig. 7a and b), which is in agreement with previous studies using grapevine dryness indices for Europe (Fraga *et al*., 2013). This intensification is expected to be stronger in regions where there are currently lower water stress conditions, that is in central and northern Europe. While in these regions precipitation is indeed expected to increase, the projected warming and enhanced evapotranspiration should have a strong negative effect on plant water status. In southern Europe, where water stress is already moderate, its strengthening is less apparent (Fig. 7b), despite the severe water stress levels in future (e.g. southern Iberia).

Regarding the nitrogen demand, it is clear that it should increase throughout Europe, with the exception of southern Iberia and regions in Italy and along the Aegean Sea. The decrease in nitrogen demand may be tied to severe water stress levels in these regions, limiting plant growth, thus decreasing the demand for this mineral nutrient. In these regions, nitrogen saturation in the plant could be achieved sooner, meaning that a nitrogen increase may not necessarily produce higher yields, owing to low water availability.

Discussion

The present study is focused on the application of the STICS crop model to assess present (1980–2005) and future (2041–2070) mean suitability for grapevine growth, yield, phenology and water/nitrogen stress conditions over Europe. For this purpose, STICS was applied to grapevines and coupled with state-of-the-art climate, soil and terrain data sets. Several grapevine variables simulated by STICS were compared to observational data sets throughout Europe. Although some model bias was identified, the simulated variables depict a high agreement with observed phenology, LAI, yield and water/nitrogen stress conditions. The identified bias can be attributed to the large extent of the target area and to some corresponding assumptions in model parameterizations, such as regional viticultural practices and varieties. Nonetheless, the different contrasts in European viticulture were skilfully modelled by STICS, such as the differences in yield and phenological timing between warmer (southern Europe) and cooler regions (central and northern regions). Furthermore, the model provides a realistic viticultural zoning in Europe, when compared to recent land cover data (EEA, 2002).

Applied to climate change scenarios, the crop model projections indicate that viticulture may undergo many changes until the end of the 21st century. The first noticeable change is the increase in potential areas suitable for viticulture in Europe. Currently, viticulture is mostly limited to areas below 50°N. In the future, however, viticulture may expand to areas up to 55°N, effectively increasing the potential grapevine growing areas. Nonetheless, the current European regulatory enforcements and the interest in other crops (e.g. wheat, barley, maize) may limit this expansion (Ingram & Porter, 2015). For southern European winemaking regions (i.e. Italy, Spain and Portugal), projections indicate viticulture will still remain suitable, although its sustainability will decrease (lower yields) due to enhanced warming

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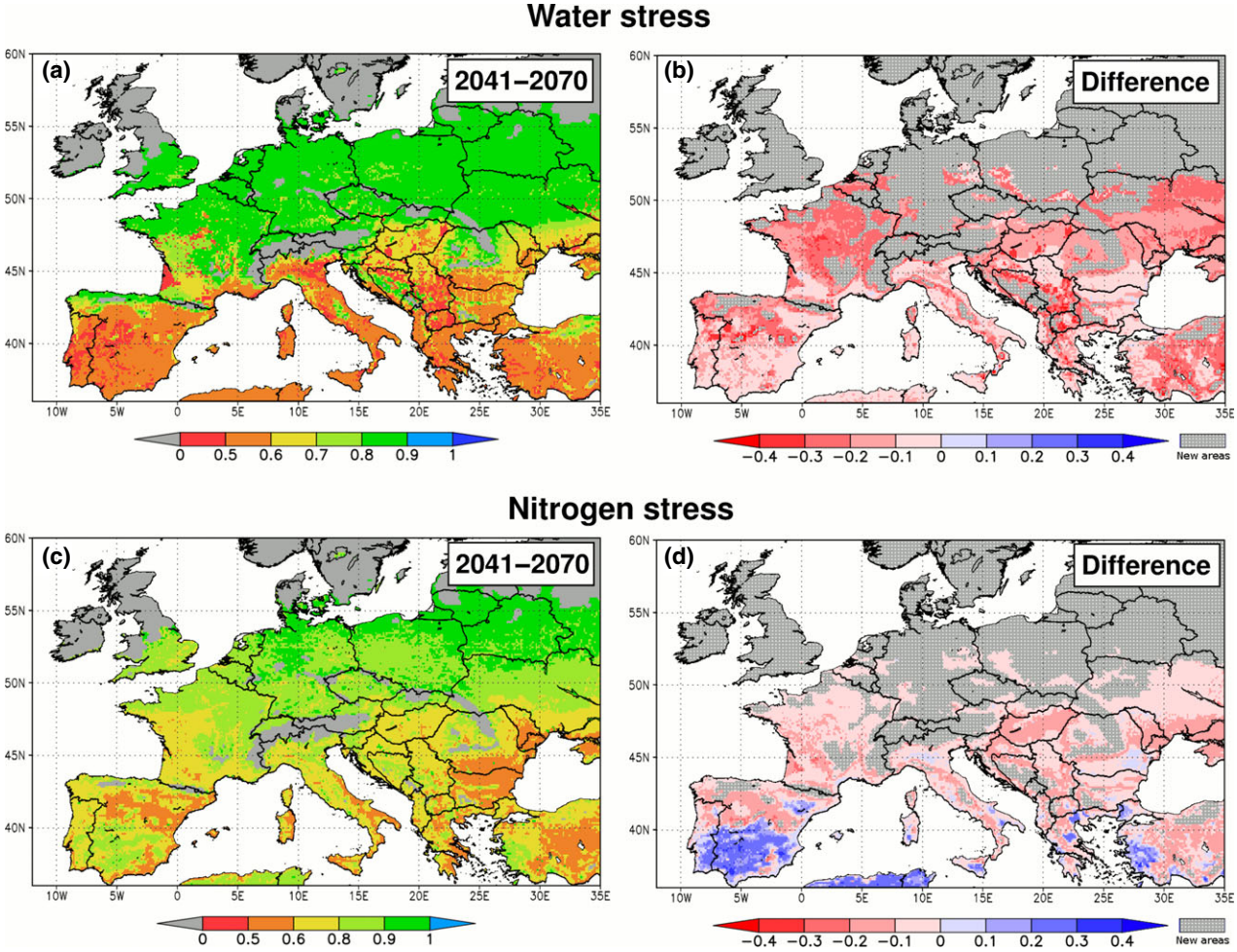


Fig. 7 Simulateur mulTIdisciplinaire pour les Cultures Standard (STICS) model simulations for the (a) water and (c) nitrogen stress over Europe (2041–2070), along with the respective differences to the recent past period (b, d). The areas where STICS provides no yield outputs (0 t ha^{-1}) are shown in grey.

and drying conditions (Moriondo *et al.*, 2013; Toth & Vegvari, 2016).

With respect to phenology, projections point at advancements in the timings of each event. Additionally, spatial differences in the anticipation of budburst and harvest may lead to heterogeneous changes in the length of the grapevine development cycle. As phenological timings are projected to occur earlier in the year, grapevines can be further exposed to heat and water stress during summer, which may result in yield losses. These shifts towards earlier phenophase onsets can also potentially threaten the regional wine characteristics and typicity. In effect, in warmer climates, alcohol levels should be attained more rapidly, with unbalanced acidity, fundamentally affecting wine microbiological risks and wine conservation potential (Orduna, 2010). These anticipations can also have strong impacts on management activities, such as spray application against pests/diseases, as these interventions usually depend on the phenological stages.

Projections also highlight that most viticultural regions may have strengthened water stress under future conditions, mostly due to the increase in evapotranspiration. The effect of the enhanced CO_2 concentration on grapevine physiology simulated by the model partially offsets the negative effects from dryness. This effect should be strongest over central/northern Europe, where projections suggest dryness not reaching severe levels. Conversely, regions in southern Europe may have to deal with excessive dryness, which should be a major limiting factor of growth and yield. Nitrogen demands are expect to also increase over central/northern Europe to cope with the enhanced biomass growth (LAI) triggered by higher CO_2 concentrations. This relationship between CO_2 and nitrogen has been previously reported (Soussana *et al.*, 2005). This may have strong

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impacts on ecological/hydrological systems, as fertilization with added N may be a major source of global pollution (e.g. groundwater contamination). In the future, maintaining yields while reducing the impact of nitrogen pollution can be an additional challenge for winegrowers in these regions. As such, agronomic research should help winegrowers improve field practices and optimize nitrogen use to improve soil quality. Over southern Europe, however, nitrogen demand should diminish, due to a decrease in biomass. Despite the negative impacts, this may represent an indirect benefit for the sustainability of local ecosystems.

Our results suggest that grapevine yields in southern Europe may decrease with increasing dryness. In fact, water stress is expected to have the strongest negative impact on yields, in opposition to other positive effects (higher CO_2). Over central and northern winemaking regions, the increase in CO_2 may partially compensate for increased dryness, resulting in higher yields. This compensation effect has already been found in several studies (Bindi *et al*., 1996; Kizildeniz *et al*., 2015). The impacts are tied not only to yields, but also to quality. While increased sugar/alcohol levels may benefit regions over northern Europe, faster ripening during the warmest part of the year over southern Europe is expected, being unfavourable to quality.

Given the results shown in the current study, an adequate and timely planning of suitable adaptation measures needs to be adopted by the winemaking sector. It is clear that the current relationship between wine quality and production could be challenged in the future. For southern European winemaking regions, given the projected yield reduction, standard table wines may struggle to remain economically viable. On the other hand, many vineyards are already targeted at making premium wines and growers tend to prefer low-yield vineyards (1–3 t ha^{-1}) (Santos *et al*., 2011). Therefore, it is reasonable to believe that, as long as there is market demand for these wine types, viticulture may remain sustainable at low yields.

One of the main adaptation measures to future drier climates in southern Europe is the improvement of grapevine water use efficiency. Water scarcity and competition may become one of the main problems in some areas and smart irrigation strategies should also be considered, when economically viable and environmentally sustainable (Chaves *et al*., 2007). More adapted varieties/rootstocks or training systems to enhanced stress conditions may also be considered (Jones, 2006). Soil tillage or cover cropping may also potentially increase water use efficiency and soil moisture (Bahar & Yasasin, 2010). Although the adoption of these measures may change the wine characteristics and the *terroir* of each viticultural region in Europe, they should be explored in each vineyard taking into account its specificities, as they might be required to warrant the future sustainability of the winemaking sector.

The current study is a first approach using a crop model at such a large-scale level (Europe). As such, it becomes necessary to continue to test, evaluate and improve such tools to provide more accurate information regarding climate impacts on viticulture and the specific adaptation measures to climate change. Furthermore, the simulation of the effects of extreme weather events on grapevine yield and quality attributes should also be analysed in future research, as it may provide valuable awareness on climate change impacts on viticulture.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Main model components in STICS and methods of calculation following Moriondo *et al*. (2015).
Figure S1. Location of the points used for the full 26-year model runs (from 1980 to 2005). The selected points correspond to widely known viticultural regions in Europe.
Figure S2. Comparison of STICS outputs for the selected points in Europe (Fig. S1): In the *x* axis, daily climatic normals (1980–2005) where used as STICS inputs, whereas in the y axis, individual daily annual climatic data was used as STICS inputs and the results were averaged afterwards.
Figure S3. Differences between future RCP8.5 (2041–2070) and recent-past (1980–2005) (a) annual mean daily temperature (°C) and (b) annual mean daily precipitation (mm).
Figure S4. Differences between future (2041–2070) and recent-past (1980–2005) phenological intervals (a) budburst–flowering, (b) flowering–veraison, (c) veraison–maturation and (d) budburst–maturation.
Figure S5. STICS model simulations for budburst, flowering, veraison and maturation over Europe under RCP4.5 and 8.5.
Figure S6. STICS model simulations for Leaf area index and yield over Europe under RCP4.5 and 8.5.

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